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SOURCE: MEPC.209(63).pdf
SEDIMENT CONTROL ON SHIPS (G12)

1 PURPOSE

1.1 Regulation B-5.2 of the Convention requires that ships described in regulations B-3.3 to B-3.5 should, without compromising safety or operational efficiency, be designed and constructed with a view to minimize the uptake and undesirable entrapment of sediments, facilitate removal of sediments and provide safe access to allow for sediment removal and sampling, taking into account these Guidelines. Ships described in regulation B-3.1 of the Convention should, to the extent practicable, also comply with regulation B-5.2, taking into account these Guidelines.

1.2 The purpose of these Guidelines is to provide guidance to ship designers, shipbuilders, owners and operators in the devel opment of ship structures and equipment to achieve the objectives of paragraph 1.1 and, t hereby, reduce the possibility of introducing harmful aquatic organisms and pathogens.

1.3 There may be a conflict between preventing accumulation of sediments and preventing the discharge of harmful aquatic organisms and pathogens.

2 INTRODUCTION

2.1 Water taken up as ships' ballast can contain solid alluvial matter that, once the water

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preventing the discharge of harmful aquatic organisms and pathogens.

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2.1 Water taken up as ships' ballast can contain solid alluvial matter that, once the water is becalmed in a ship's ballast tank, will settle out onto the bottom of the tank and other internal structures.

2.2 Aquatic organisms can also settle out of the ballast water and can continue to exist within the sediment. These organisms can surviv e for long periods after the water they were originally in has been discharged. They may thereby be transported from their natural habitat and discharged in another port or area where they may cause injury or damage to the environment, human health, property and resources.

2.3 Regulation B-5.1 of the Convention requires that all ships remove and dispose of sediments from spaces designated to carry ballast water in accordance with the Ballast Water Management Plans. These Guidelines are to assist ship designers, shipbuilders, owners and operators to design ships to minimize the retention of sediment. Guidance on the management of sediment is contained in the Guidelines for ballast water management

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owners and operators to design ships to minimize the retention of sediment. Guidance on the management of sediment is contained in the Guidelines for ballast water management and development of ballast water management plans (G4).

3 DEFINITIONS

3.1 For the purposes of these Guidelines, the definitions in the International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004 (the Convention) apply.

3.2 Ballast water tank For the purposes of these Guidelines, a ballast water tank is any tank, hold or space used for the carriage of ballast water as defined in Article 1 of the Convention.

4 DESIGN FOR REDUCING ACCUMULATION OF SEDIMENT

4.1 Ballast water tanks and their internal structure should be designed to avoid the accumulation of sediment in a ballast tank. The following should, as far as is practicable, be taken into account when designing ballast tanks:

.1 horizontal surfaces to be avoided wherever possible;

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accumulation of sediment in a ballast tank. The following should, as far as is practicable, be taken into account when designing ballast tanks:

.1 horizontal surfaces to be avoided wherever possible;

.2 where longitudinals are fitted with face bar stiffeners, consideration should be given to fit the face bar stiffeners below the horizontal surfaces to aid drain off from the stiffeners;

.3 arrange for induced flows of water, either by pump forces or gravitational forces, to wash along horizontal or near horizontal surfaces so that it re-suspends already settled sediment;

.4 where horizontal stringers or webs are required, drainage holes to be as large as possible, especially if edge toe-stops are fitted where horizontal stringers are used as walkways, to encourage rapid flow of water off them as the water level in the tank falls;

.5 internal girders, longitudinals, stiffeners, intercostals and floors, where fitted, should incorporate extra drain hol es which allow water to flow with

minimal restriction during discharge and stripping operations;

.6 where inner members butt against bulkheads, their installation should be

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fitted, should incorporate extra drain hol es which allow water to flow with
minimal restriction during discharge and stripping operations;

.6 where inner members butt against bulkheads, their installation should be
such as to prevent the formation of stagnant pools or sediment traps;

.7 scallops should be located at the joints of the inner bottom (tank top)
longitudinals or intercostals and floors to allow for good airflow, and thus
drying out of an empty tank. This will also allow air to escape to the air pipe
during filling so that minimum air is trapped within the tank;

.8 pipeline systems should be designed such that, when deballasting,
disturbance of the water in the tank is as powerful as possible, so that the
turbulence re-suspends sediment; and

.9 flow patterns in ballast water tanks should be studied (for example by the
use of Computational Fluid Dynamics (CFD)) and considered, so that
internal structure can be designed to provide effective flushing. The amount
of internal structure in double bottom tanks will reduce the scope for
improving flow patterns. The hydrodynamic performance of the ballast tank
is crucial to ensure sediment scouring.

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improving flow patterns. The hydrodynamic performance of the ballast tank
is crucial to ensure sediment scouring.

4.2 Any designs depending upon water flow to re-suspend sediment should, as far as possible, be independent of human intervention, in order that the workload of ships' crews is minimal when operating the system.

4.3 The benefits of design concepts for reducing sediment accumulation are that there is likely to be good sediment removal while deballasting, with minimum retention of sediment in the tanks, and therefore a reduction or no need for removal by other means.

4.4 The design of all ships should provide safe access to allow for sediment removal and sampling.

4.5 The design of ballast water systems should, as far as practicable, facilitate installation of high sea suction points on each side of the ship.

4.6 When practical, equipment to remove suspended matter at the point of uptake

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4.6 When practical, equipment to remove suspended matter at the point of uptake should be installed.

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